

Zomato Restaurant Analytics Dashboard

1. Project Overview

The Zomato Restaurant Analytics Dashboard is a data analytics project developed using Python and Power BI. The objective of the project is to analyze restaurant data and identify trends related to customer ratings, pricing, cuisines, restaurant types, online ordering, and table booking services.

2. Dataset Description

Dataset Size:

- Total Records: 7,105

After Data Cleaning:

- Total Records: 6,984

Features Used:

- Restaurant Name
- Restaurant Type
- Rate (Out of 5)
- Number of Ratings
- Average Cost (Two People)
- Online Order
- Table Booking
- Cuisines Type
- Area
- Local Address

3. Data Cleaning

The following preprocessing steps were performed:

- Removed unnecessary columns:
 - Unnamed: 0
 - Unnamed: 0.1
- Removed missing values from:
 - Rate (Out of 5)
 - Average Cost (Two People)

- Checked and removed duplicate records.
- Prepared a clean dataset for analysis and visualization.

Final Dataset Size:

- 6,984 Records
- 10 Features

4. Exploratory Data Analysis

The following analyses were performed:

- Restaurant Type Analysis
- Online Order Analysis
- Table Booking Analysis
- Area Analysis
- Cuisine Analysis
- Rating Distribution Analysis
- Cost Distribution Analysis
- Cost vs Rating Analysis

A custom metric called Popularity Score was created:

Popularity Score = Rating × Number of Ratings

This metric was used to identify the most influential restaurants.

5. Dashboard Development

The dashboard was developed using Power BI and includes:

KPIs:

- Total Restaurants
- Average Rating
- Average Cost
- Online Order Restaurants
- Table Booking Restaurants

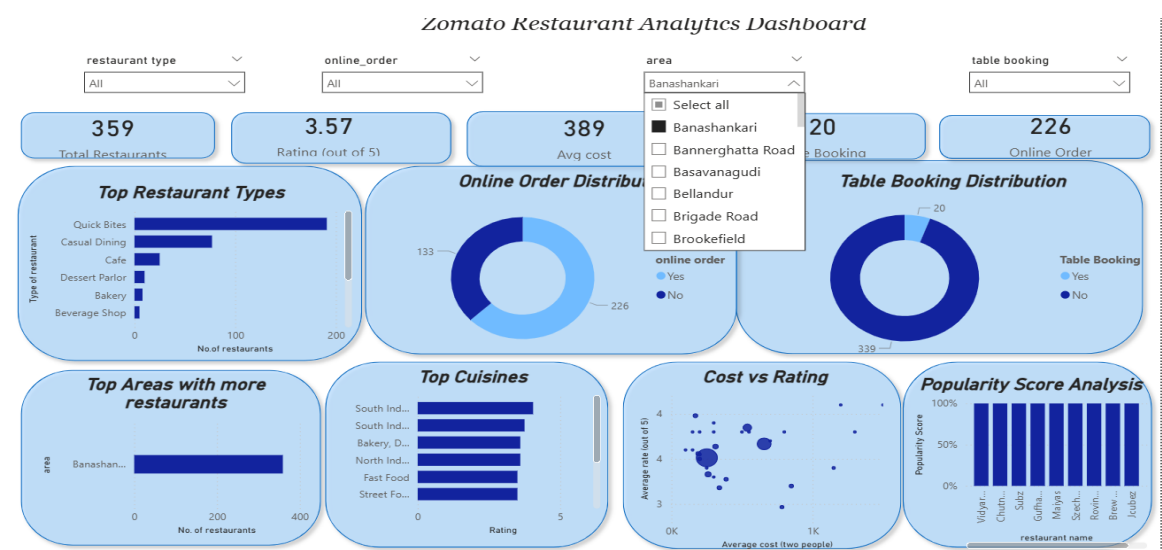
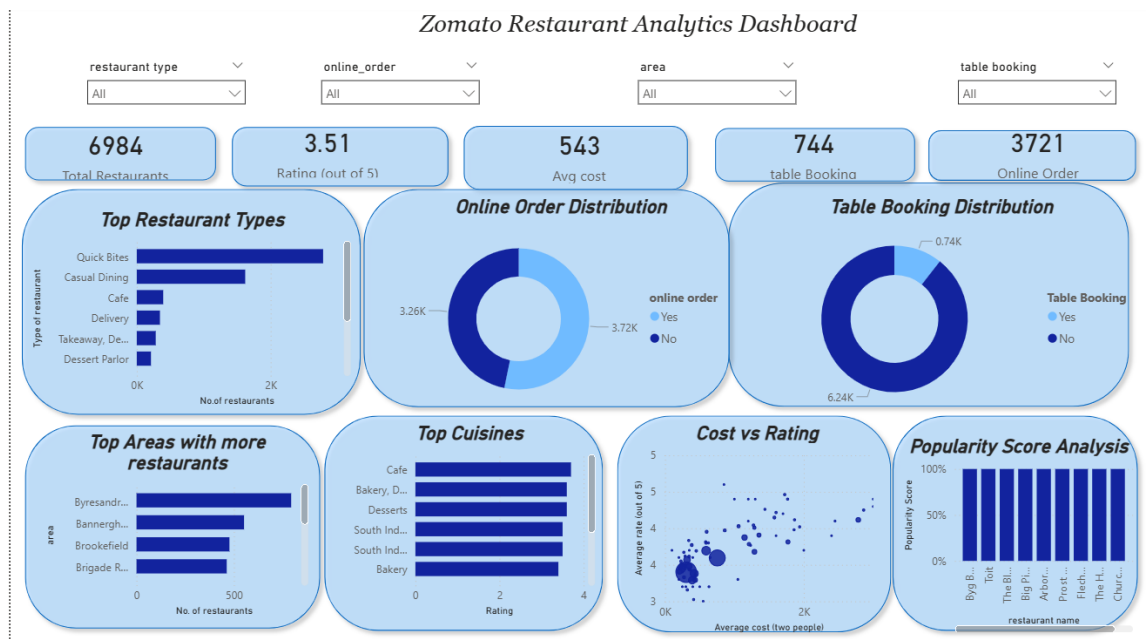
Visualizations:

- Top Restaurant Types
- Online Order Distribution
- Table Booking Distribution
- Top Areas

- Top Cuisines
- Cost vs Rating Analysis
- Top Restaurants by Popularity Score
- Cuisine-wise Average Rating

Interactive slicers were added for:

- Area
- Restaurant Type
- Online Order
- Table Booking



6. Key Findings

- Quick Bites is the most common restaurant type.
- More than 53% of restaurants offer online ordering.
- Restaurants with table booking have significantly higher ratings (4.04) compared to restaurants without table booking (3.45).
- The average restaurant rating is 3.51 out of 5.
- The average cost for two people is ₹543.
- Byresandra–Tavarekere–Madiwala is the area with the highest concentration of restaurants.
- Byg Brewski Brewing Company and Toit are among the most popular restaurants based on the Popularity Score.

7. Conclusion

The project successfully analyzed restaurant trends using Python and Power BI. The findings provide insights into customer preferences, restaurant performance, pricing patterns, and service offerings. The dashboard enables users to explore restaurant data interactively and supports data-driven decision-making.